

4.1 ServiceNow Capstone Documentation

Project: CampusAssist AI – Smart IT Support Portal and AI-Assisted Troubleshooting System in ServiceNow
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1. Project Overview & Significance

CampusAssist AI is a comprehensive ServiceNow-based solution designed to transform campus IT support. It addresses the common pain points of delayed ticketing and limited accessibility by providing a 24/7 intelligent self-service portal. The goal is to offload routine hardware and software queries, allowing technical staff to focus on high-priority infrastructure projects.

2. Technical Architecture & Logic

The system utilizes a **Scoped Application** framework to ensure modularity and security. Central to the logic is the **CampusCare AI Chatbot**, which employs state-machine logic to guide students through troubleshooting. The data flow is governed by ServiceNow's **Glide APIs**, ensuring real-time form validation and secure record creation.

3. Technology Used

- ServiceNow, Scoped Application, Service Portal, Service Portal Widgets
- HTML, CSS, JavaScript, AngularJS, Glide APIs

4. ServiceNow Application Details

The implementation involved configuring custom Service Portal widgets and defining Access Control Lists (ACLs) to secure student data.

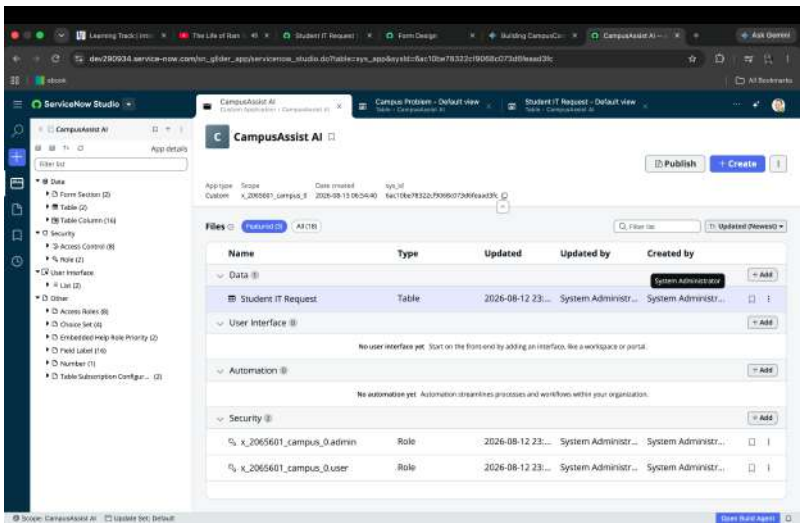


Figure 1 – Application Scope, Files, and ACL Settings

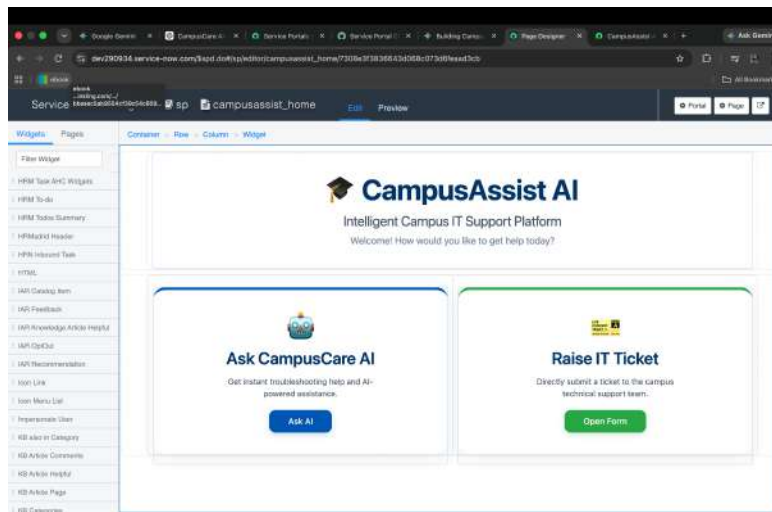


Figure 2 – CampusAssist AI Service Portal Homepage

5. Data Schema & Table Structure

We implemented a custom table `x_2065601_campus_0_student_it_request` to store comprehensive student information alongside their reported issues. This table is optimized for reporting and analytics.

Field	Type	Purpose
student_name	String	Identity
student_id	String	Tracking
department	Choice	Categorization
issue_type	Choice	Categorization
description	String	Details
state	Choice	Lifecycle Status

Column label	Column name	Type	Reference	Max length	Default value	Index
Affected Equipment	affected_equipment	String		40		
Created	sys_created_on	DateTime				
Created By	sys_created_by	String		40		
Department	department	Choice				
Description	description	String		40		
Email	email	String		40		
Issue Type	issue_type	Choice				
Location	location	String		40		
Phone Number	phone_number	String		40		
Related Problem	related_problem	Reference	Campus Problem			
Student Affected	student_affected	String		40		
Student ID	student_id	String		40		

Figure 3 – Detailed Dictionary and Table Structure

6. CampusCare AI & Guided Troubleshooting

CampusCare AI is an interactive chatbot that provides guided troubleshooting for common student IT issues such as WiFi, hardware, software, and account lock problems through structured questions and step-by-step instructions.

7. Multiple Issue Handling

After one issue is completed, the conversation state is reset so the student can start another issue without the previous issue affecting the new conversation session.

8. Escalation & Workflow Logic

If the AI troubleshooting fails to resolve the issue, the workflow automatically transitions the user to a structured request form. UI Policies were applied to mandate essential fields, ensuring IT support receives all required information at the point of submission.

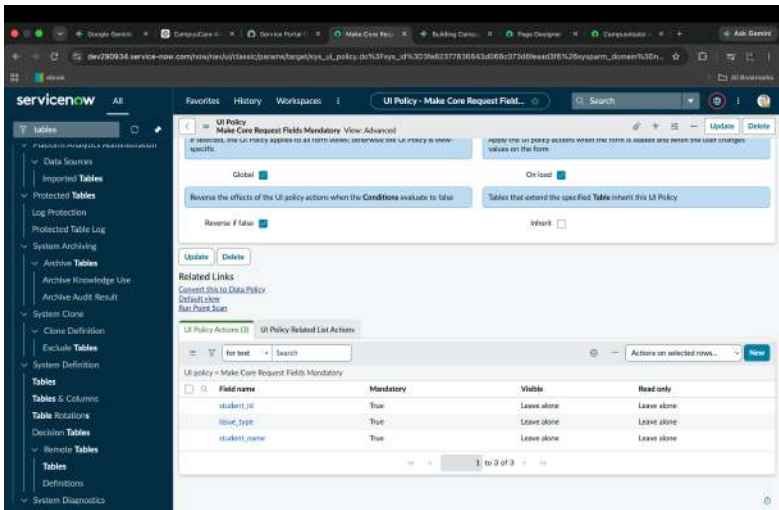


Figure 4 – Escalation Workflow & Request Form UI Policies

9. System Architecture Flow

Student → CampusAssist AI Portal → CampusCare AI → Troubleshooting → Resolved / Escalate → ServiceNow Request Form → Student IT Request Table → IT Support

10. Implementation & Validation Testing

Thorough testing was conducted, including: 1) Portal UI functionality; 2) Chatbot conversational logic reset; 3) Form submission error handling; 4) Security/permission validation for student records.

The screenshot displays the ServiceNow interface for a 'Student IT Request'. The left sidebar shows the 'Tables' menu with various options like 'Platform Analytics Administration', 'Data Sources', 'Imported Tables', 'Protected Tables', 'Log Protection', 'Protected Table Log', 'System Archiving', 'Archive Tables', 'Archive Knowledge Line', 'Archive Audit Result', 'System Clone', 'Clone Definition', 'Exclude Tables', 'System Definition', 'Table', 'Table & Columns', 'Table Relations', and 'Decision Tables'. The main form area is titled 'Student IT Request' and 'Create Created'. It contains several input fields: 'Student Name', 'Student ID', 'Department' (set to 'CSE'), 'Email' (with a red error message: 'Please enter a valid email address (e.g., student@campus.edu)'), 'Phone Number', and 'Related Problem'. Below these fields are two tabs: 'Issue Information' and 'Support Details'. The 'Support Details' tab is active, showing fields for 'Location' (set to 'Hobart Block A'), 'Issue Type' (set to 'WIFI Issue'), 'Description', 'Affected Equipment', and 'Student Affected'. A 'Submit' button is located at the bottom of the form.

Figure 5 – Testing Scenarios & UI Policy Validation Results

11. Project Result

Successfully established the CampusAssist AI scoped application within ServiceNow, featuring a dedicated Service Portal homepage, custom request tables, automated UI validation policies, and guided troubleshooting pathways for campus students.

12. Future Improvements

- Knowledge Base integration & automated notifications
- Analytics dashboards and natural-language AI capabilities
- Mobile support optimization

13. Conclusion

CampusAssist AI successfully streamlines campus IT support by offering students an intuitive self-service portal and guided chatbot troubleshooting experience, effectively reducing administrative burden while ensuring seamless ticket escalation.

14. Supporting Technical Documentation

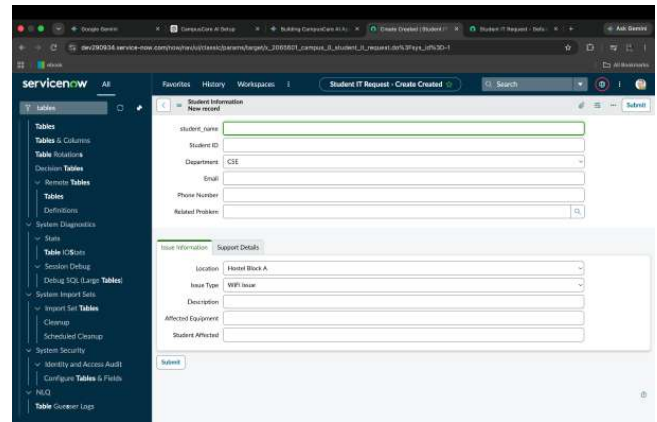
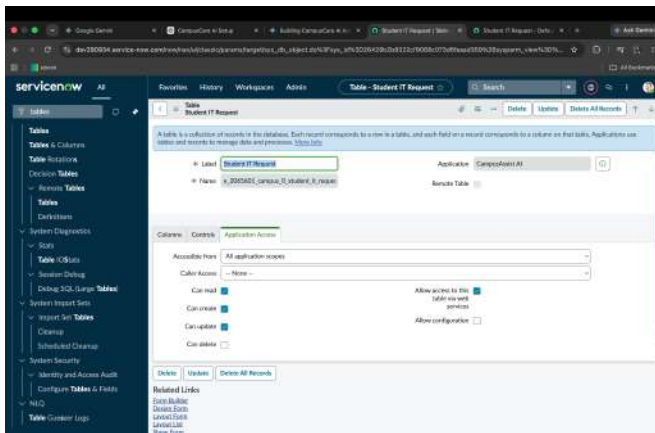


Figure 6 – Access Settings & Table Security

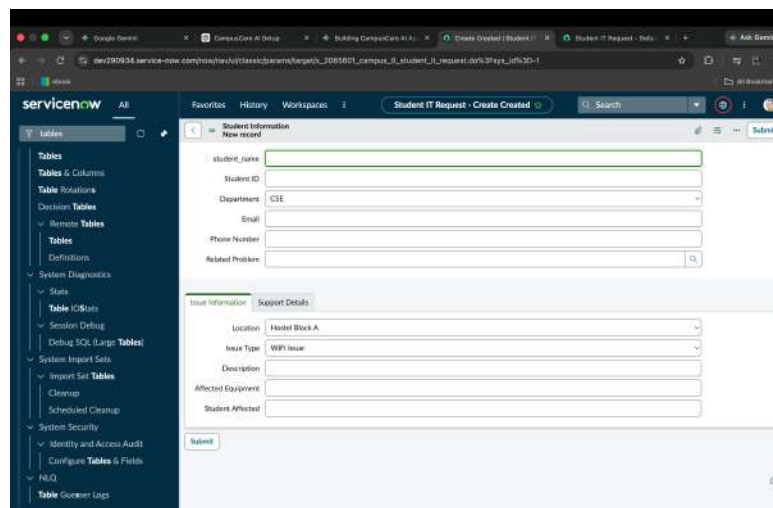
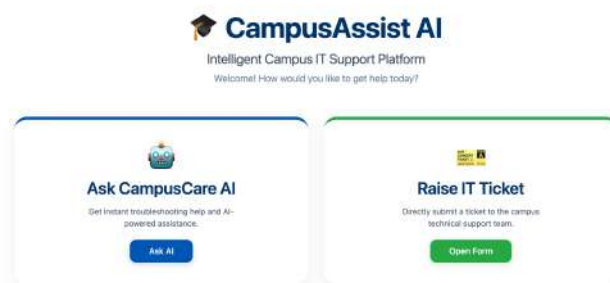


Figure 7 – Student Information Dictionary Form Definitions



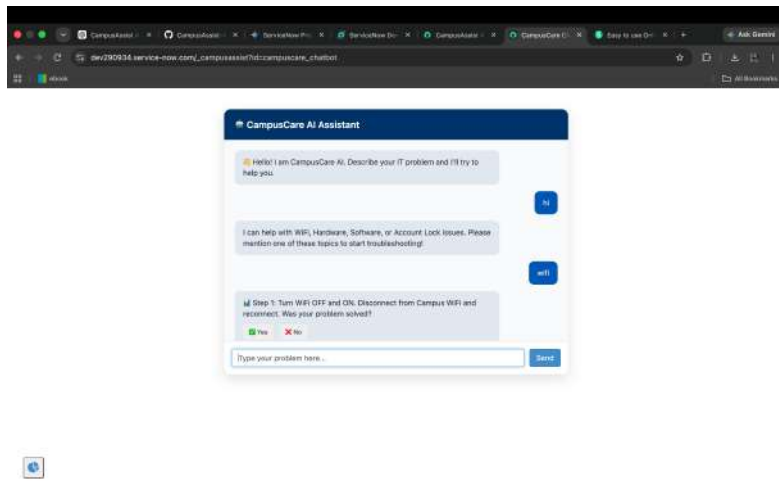


Figure 8 – Final Form Submission Error Handling